

Lab Exercise 1: Introduction to C++

Submit as .cpp files.

 Problem 1: Greeting with Default & Overloaded Functions**Task:**

Write two overloaded functions named greet:

1. `greet(string name)` → prints "Hello, [name]!"
2. `greet(string name, string greeting)` → prints "[greeting], [name]!"

Also, write a third version:

3. `greet(string name = "Guest")` → uses a **default argument**

Test in main():

Cpp:

```
greet("Ali");           // Hello, Ali!
greet("Sara", "Hi");    // Hi, Sara!
greet();                // Hello, Guest!
```

 **Focus:** Function overloading + default arguments

 Problem 2: Bitwise Power-of-Two Checker**Task:**

Write a function `bool isPowerOfTwo(int n)` that returns true if `n` is a power of two (e.g., 1, 2, 4, 8, 16...), else false.

Hint: Use bitwise AND (&).

 **Trick:** A number `n` is a power of two **iff** `n > 0` and `(n & (n - 1)) == 0`.

Sample:

Cpp:

```
isPowerOfTwo(8); // true
isPowerOfTwo(6); // false
```

 **Focus:** Bitwise operators, logical thinking

✅ Problem 3: Swap Using References

Task:

Write a function `void swap(int &a, int &b)` that swaps two integers **using references** (not pointers!).

In `main()`, read two numbers from the user, call `swap`, and print them.

Sample Input/Output:

Enter two numbers: 5 10

After swap: 10 5

 **Focus:** References vs pointers, clean C++ style

✅ Problem 4: Dynamic Array Sum

Task:

Ask the user for `n`, then dynamically allocate an array of `n` integers using `new`.

Read `n` numbers into the array, compute their sum, and print it.

Don't forget to `delete[]` the array!

Sample:

How many numbers? 3

Enter numbers: 4 7 2

Sum = 13

 **Focus:** Dynamic memory (`new/delete`), basic I/O

✅ Problem 5 (Challenge): Fibonacci with Memoization (Preview of DP)

Task:

Write a recursive function `int fib(int n)` that computes the `n`th Fibonacci number **but stores previously computed values in a global array** (size 100, initialized to -1).

This is a **top-down DP (memoization)** intro!

Rules:

- If `fib_cache[n] != -1`, return it.
- Else, compute it, store it, then return.

Test:

Cpp:

```
cout << fib(10); // Should output 55 quickly, even for n=40
```

 **Focus:** *Recursion + caching (gentle intro to DP concept)*